Engineering Forensics

ENP-0112

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Divya Chandra, Andy Liu, Parisa Arastu

Agenda

- Introductions
- Course content
- Syllabus
- Generative AI discussion
- Extras
  - Websites, context, mitigations

Introductions

Lecturers

Divya Chandra, Ph.D. {d.chandra@tufts.edu}

Andrew M. Liu, Ph.D. {aliu17@tufts.edu}

Teaching Assistant

Parisa Arastu, Arts & Sciences 26

{Parisa.Arastu@tufts.edu}

Class Members

- Nate Brophy
- En Chen (G; graduate student)
- Jack Chung
- Ainsley Cui (G)
- Annica Grote
- Amanda Hasenauer (G)
- Austin Jobson
- Quian Li (G)
- Rianna Liu
- Sachita Mane (G)
- Joey Marmo
- Varun Nair (G)
- Lance Nielson-Konzen (G)
- Elsa Ostenson
- Matt Peryea (G)
- Sahir Surani (G)
- Deelia Wang (G)
- Rachel Woo
- Lucy Zimmerman
- *Adin Handler? (not in SIS)*

Engineering Forensics?

- Common usage…..Forensic science applies scientific methods and processes to solve crimes or identify liability
  - Tort lawyers, expert witnesses and forensic science
  - Structural failures, engineering design deficiencies
  - Ex. CSI television crime drama

- Forensic human factors
  - Analyses of injury cases
  - Expert witness in legal cases
  - Person-product or person-environment interaction –severe injury or death

- This course uses “forensic” in an analytic manner
  - Learn from failures
  - Determine what went wrong and why
  - Identify mitigations

Course Goals

- Develop skills and knowledge needed to strategically **investigate and analyze** human-centered safety issues that can lead to unexpected adverse consequences in operational environments.
- Practice **explaining the causes of incidents** concisely and clearly, in both oral and written formats, using accurate and standard human factors terms.
- Practice **explaining recommended counter measures and corrective actions**, in both oral and written formats.
- **Evaluate tradeoffs in system design** that can affect safety.
- Participate in class discussions.
- Give examples of human factors issues that affect safety.

Course Elements

Issues/theory ("Sidebars")

Cases

Analysis Methods and Frameworks

Mitigations

Example “Sidebars”

After the talks, instructors will dive into issues and theories during the sidebar lectures. These will cover theory and context, or other related information.

Example topics:

- Attention and Monitoring
- Automation, trust
- Decision making, complacency
- Distraction/inattention
- Fatigue
- Usability
- Warnings/alarms
- Organizational culture

Course Structure

- Typical class begins with the student presentations, then sidebar
  - There will guest lectures, without student talks.

- Grading – emphasizes original work
  - Reading responses (15%)
    - Weekly, submitted through Rumi plug-in on Canvas
  - Presentations (45%)
    - Two individual oral presentations, focused on specific aspects of a real case
    - Rubric considers content, delivery, and handling questions
  - Exams (40%)
    - Written mid-term and final exams, in-class and on-line components

- Ungraded but important to developing knowledge and skills
  - Attendance/Participation in discussions and Q&A for student presentations

Submission Due Dates

- Reading responses usually due Wed midnight (confirm in Canvas)
  - Readings will be available throughout the term, even if unread.
  - No late responses will be accepted for credit.
- Presentation schedule will be developed by TA and posted to Canvas home page.
  - Topics for each talk will be finalized at least 2 weeks before the presentation date.
- Midterm exam on March 9 during class
- Final exam – May 1, 12-2p (based on Block Schedule E/E+)
- **Tell the TA and instructors as early as possible if you have any problems with the exam or presentation dates, or if you will be out on known dates.**
- We will also schedule a trip to the USDOT Volpe Center human factors laboratory.
  - Attendance is optional but recommended if possible.

Reading Tips

- GenAI summaries can be helpful, but can also be generic and uninteresting
- Read the Abstracts and Executive Summaries in full
- Skim the table of contents. What looks interesting to you?

➢ Goal is to become familiar with the subject, not necessarily an expert on it.

Reading Responses

- Approximately 20 readings over the term
- Responses should be short and in personal writing style, not formal.
- Each reading response counts for 1% of grade up to 15%
  - Up to 5% extra credit for additional reading responses
- Rumi plug-in
  - Simple word processor, configured to record all entries including use of copy/paste and AI tools to encourage original work.
  - Practice using Rumi for reading responses; we may use it for exams as well.
  - https://sites.tufts.edu/ets/tools/rumi/

Rumi

Rumi is a writing tool which makes the writing process visible. Integrated into Canvas, it deters AI misuse by encouraging students to write in their own voice and do their own thinking. It will try and detect when students are not creating their own writing, and collects evidence you can use to determine if the student is following your AI use policy.

Rumi does this by:

1.Recording the students writing so that you can evaluate their process. When students create their own work, it looks very different from when they transcribe from AI or elsewhere.

2.Preventing students from copying and pasting from other sources without attribution. Whenever a student pastes text into the assignment, Rumi asks for a justification of where the text came from. You can evaluate if this is a quote, draft, or chunk of text taken from AI.

3.Evaluating the student's writing behavior. Rumi has a number of metrics that help you see, at a high level, which submissions need to be evaluated for unethical AI use.

4.Allowing students to use AI tools that you have approved and to records those interactions for you to review.

Analysis of a Case-High Level

- Describe the facts of the situation
- What are the causes?
- What should be done?

Warning

- Situations and videos can be alarming
- Focus on series of missteps leading to the incident, not the human pain and suffering
- Try to humanize, but not personalize

How to begin…

- The best sources are the original accident investigation reports.
  - US National Transportation Safety Board (NTSB) has well written, plain language reports.
  - Read the Executive summaries for an overview, then decide which sections to dive into further for your particular area of interest ("lens" that you are applying)
- Higher level summaries are often available online, for example:
  - https://lessonslearned.faa.gov/general.cfm
  - Wikipedia

Note: GenAI tools can be used to search and summarize references (if confirmed that they exist), but the presentation content should not be created with AI.

Presenting the Case – General Example

1. What happened? (Video clips or simulations can help.)
2. Provide historical perspective on the operation in which the accident occurred
3. Timeline: Present a graphic timeline showing when various factors occurred.
4. Describe the human/operator behavior in the situation
5. What were the consequences (injuries, fatalities, economic damage)?
6. Characterize the culture of the organization
7. Explain the primary and contributing factors
8. Measures that should have been taken to avoid the accident or mitigate its consequences?
9. Policy recommendations: Develop and present recommendations for changes to reduce likelihood of recurrence

Presentation Rubric

- Overall Grade, Max 22.5 points for each (45 points total)
- <u>Content</u>
  - Problem Statement (4 points)
    - Summary of case background and significance related to topic.
    - Demonstrated understanding of the context of the case.
  - Analysis and Discussion (4 points)
    - Analysis of situation, comparison to theory and/or previous cases, clear description and discussion of human-related issues.
  - Conclusions. (4 points)
    - Presentation of conclusions with important results highlighted.
- <u>Delivery</u>
  - Interaction and Voice. Eye contact and engagement with audience. Appropriate voice volume, clarity, pace, and inflection. (4 points)
- <u>Questions/Answers (Q/A)</u>
  - Ability to respond to questions. (6.5 points)

Mid-term and Final Exam Formats

- New format this year, trial.
  - In-person, during class for part of the exam. Also one online essay component for each exam.
  - May use Rumi plug-in to avoid hand-written responses.
  - May make adjustments for the final exam after the mid-term exam.
- If you have a letter of accommodation (e.g., more time on exams), let us know ASAP so we can make arrangements
- In-class exam components:
  1. Prepare a reaction to new reading material.
  2. Semi-prepared essay. We will give you a set of questions in advance. Will select one question from the set for the exam.
     - Can use GenAI to study and prepare
     - Suggested method is to outline a response for every question and then just write out the response during the exam time.
     - Length of response is less important than prioritization and analysis.
     - Add specific examples to support your points.
- On-line component
  - Expect one short essay with a word-limit.
  - Should refer to reading materials.
- Suggestions? Comments?

Discussion of GenAI Responses

- What is good?
- What could be improved?
- Is there anything false in the response?
- Improving the prompts?
- How can *you* “add value” to the AI response?

➢ Questions/discussion about acceptable use of AI?

Extras

Transportation Sources on the Internet

- https://www.standardsportal.org/usa_en/resources/sdo.aspx
- https://www.nhtsa.gov/laws-regulations/fmvss
- https://www.nhtsa.gov/technology-innovation/automated-vehicles-safety
- https://www.faa.gov/regulations_policies/
- https://www.fmcsa.dot.gov/regulations
- https://www.fra.dot.gov/Page/P0020
- https://www.fra.dot.gov/Page/P0010
- https://www.fhwa.dot.gov/resources/
- https://www.uscg.mil/Resources/Library/
- https://apps.dtic.mil/dtic/tr/fulltext/u2/a281401.pdf (Mil Std 1472)
- https://ntsb.gov/Pages/default.aspx
- https://www.transportation.gov/
- https://www.transit.dot.gov/
- http://www.trb.org/Publications/Publications.aspx
- https://trid.trb.org/
- http://www.tsb.gc.ca/eng/
- https://www.iihs.org/
- https://lessonslearned.faa.gov/general.cfm
- https://www.askthepilot.com/
- http://www.volpe.dot.gov
- http://zoox.com
- https://visionzeronetwork.org/
- http://www.umtri.umich.edu/
- https://aaafoundation.org/
- https://www.vtti.vt.edu/
- http://enotrans.org
- https://cars.stanford.edu/
- https://trb.secure-platform.com/a/page/AutomatedRoadTransportationSymposium
- https://www.csail.mit.edu/research/self-driving-vehicles
- http://Autosafety.org
- http://pavecampaign.org
- http://driverless.mit.edu/
- http://nsc.org
- https://www-nrd.nhtsa.dot.gov/departments/esv/27th/
- www.nfpa.org
- https://www.nist.gov/
- http://agelab.mit.edu/
- https://www.nasa.gov/centers/hq/nsc
- https://www.faasafety.gov
- https://www.youtube.com/playlist?list=PL5vHkqHi51DT2Y54kjRtmjJ3Dgaj_Sv7V
- https://www.towardzerodeaths.org/
- https://www.skybrary.aero/
- https://www.apta.com/

Some context…

Perhaps one of the more valuable results of reviewing a series of accident reports is an appreciation for the complex chain of events which often underlies an accident. This is particularly true of accidents involving human error and human factors in general. Often, the basic causes assigned to an accident are, in fact, symptoms, in contrast to root causes, a concept of considerable interest and importance. …

Technical deficiencies, when discovered, can be corrected rather easily, but most of the repeat causes include human factors related deficiencies. These are more difficult to correct, because these repeat causes are not usually root causes. For example, one of the most common and serious repeat cause factors is flying in adverse weather conditions; it is quite obvious that this is not a root cause. **A root cause has to answer the question "Why"?** (Hopf-Weichel et al., 1979, p. 5-17)

Hopf-Weichel, R., Lucaccini, L.F., Saleh, J., & Freedy, A. (1979). *Aircraft Emergency Decisions: Cognitive and Situational Variables.* Air Force Office of Scientific Research (AFOSR). Report No. AOFSR-TR-1175. https://doi.org/10.21236/ADA077413

“Mitigations” - How to address human factors issues

Individual

- Selection and training (Knowledge, Skills, and Abilities)
- Motivation, arousal (optimize attention)

Organization

- Policies and procedures (e.g., checklists)
- Safety culture

System Design

- Controls (soft or hard), “Usability” (e.g., visibility of system state, interaction with data)
- Automated systems? (take the operator out of the system)

Operational Risk

- Variation across each operation (e.g., across flights)
- Work as imagined (theory) vs. work as done (reality)